Lahore Grammar School
Johar Town Senior Boys Campus

PROJECT X

Study Guide



11 . 12 . 13 SEPTEMBER

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Team composition:

- Each team may consist of 2–4 participants.
- All members are expected to contribute fairly during the competition.
- One member must be designated as the team leader.

Robot Kit (Provided):

Every participating team will receive an identical robotics kit.
The kit includes:

- x1 Arduino Nano
- x1 Nano Wire
- x1 PCB
- x1 Motor Driver
- x4 TT Motors
- x1 Bluetooth Module
- x1 Buck Converter
- x1 Double Side Tape
- x1 4 Cell Holder
- x1 bundle Wires M-M
- x1 bundle Wires M-F
- x2 Switches
- x4 CELLS
- x4 Wheels
- x1 Chassis

No additional electronic components may be used unless provided by the organizers.

General Rules:

- Teams must use only the provided robotics kit.
- The same robot must be used throughout all rounds.
- Deliberate damage to another team's robot will result in immediate disqualification.
- Judges may inspect robots at any time during the competition.
- Teams are responsible for the safety of their robot.
- The decision of the judges will be final.

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ROUND 1:

Robot Construction Challenge:

Timings: 90 minutes shall be allotted to each team for the construction of each robot. After completion each team will get their robot verified and marked to avoid any further changes. The time allotted can be changed at the judges' discretion.

Objective: Teams will construct a fully functional robotics car using the components provided by the organizers. The objective is to assemble the robot correctly in the shortest possible time while ensuring it is fully operational.

Skills Tested:

- Mechanical Assembly
- Wiring
- Basic Electronics
- Teamwork
- Time Management
- Troubleshooting

The completed robot must:

- Be mechanically secure.
- Drive forward and backward.
- Turn left and right.
- Pass the technical inspection conducted by the judges.

Judging Criteria:

Points will be awarded for the robot to meet each of the following conditions:

Construction Time:

1. 30 minutes or less [30 Points]
2. 60 minutes or less [20 Points]
3. 90 minutes or less [10 Points]

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Judging Criteria (Continued) :
Points will be deducted for the robot if it meets the following conditions:

- Construction Time :

1. 90 minutes to 100 minutes [5 Points]
2. 100 minutes to 110 minutes [10 Points]
3. 110 minutes to 120 minutes [15 Points]

- Not Able To Pass The Following Checks Deducts 5 points each :

1. Ability to move Forward and Backwards on demand
2. Ability to turn on either sides
3. Whether the robot is mechanically stable and fully completed

More than 120 minutes will lead to disqualification. Points to be totally awarded or deducted is given in front of the criteria.

ROUND 2:
Autonomous Line Following Challenge:
In this round, teams must program their robot to navigate the official Line Following Track completely autonomously. No manual control is permitted.

Competition Format: Each team will be given 1 minute for setup, after the setup time ends they will be given 4 minutes for the official run. Failing to arrive within the setup time may lead to immediate disqualification. Timing begins once the robot crosses the start line. Timing stops once the robot reaches the finish line.

Scoring:
Teams will be ranked according to:

1. Fastest completion time.
2. Number of checkpoints successfully crossed.
3. Distance travelled (if the robot does not finish).

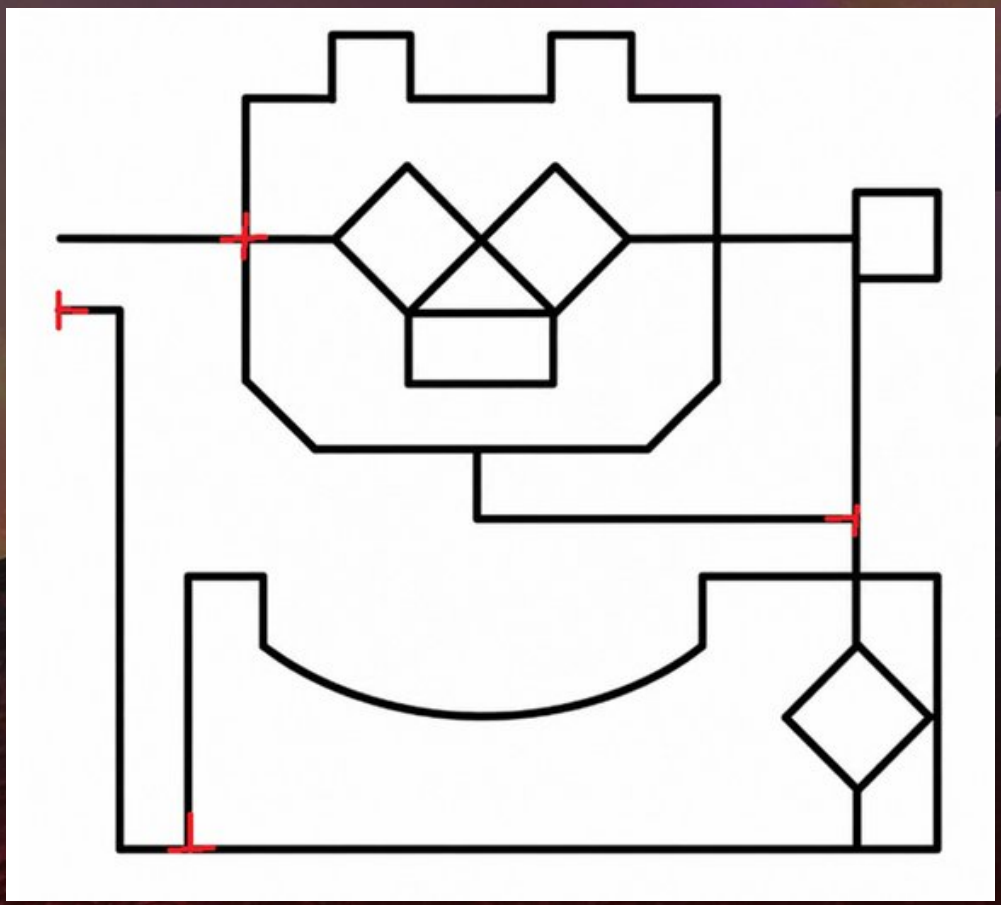
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Scoring (Continued) :
The Top 8 teams will qualify for Round 3.

Rules:

- The robot must operate fully autonomously.
- No Bluetooth or wireless control is permitted.
- Human intervention is not allowed once the run begins.
- If intervention becomes necessary, penalties may be applied at the judges' discretion.

Track Features:



Checkpoints are highlighted in red

Track Checkpoints
Marks for each checkpoint shall be as followed:

- Checkpoint 1: [10 Points]
- Checkpoint 2: [15 Points]
- Checkpoint 3: [20 Points]
- (Finish) Checkpoint 4: [25 Points]

Total : [70 Points]

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ROUND 3:

Bluetooth Flag Capture Tournament:

The final round focuses on strategy, teamwork, and driving skills. The robot built in Round 1 and used in Round 2 will now be upgraded with a **front-mounted magnet**, provided by the organizers. Robots will be controlled wirelessly using Bluetooth.

Tournament Format:

Quarterfinals:

The Top 8 teams from the combined results of Round 1 and Round 2 will be randomly paired into four 1v1 duo matches. Each match lasts 10 minutes. The winner of each match advances to the semifinals.

Semifinals:

The four winning teams will compete in two 1v1 duo matches. The winners advance to the Grand Final.

Grand Final:

The remaining two teams will compete in one final 1v1 Bluetooth Flag Capture Match in the same arena. The winner will be crowned the Project X Winner, while the losing finalist will be declared the Runner-Up.

Arena :

The arena is divided into two equal sides. Each team begins on its designated side. The two sides are connected by two bridges:

- Bridge A: Narrow Bridge
- Bridge B: Obstacle Bridge

Teams may freely choose which bridge to use throughout the match.

Flags:

Each team's side contains two types of flags.

Normal Flags Each side contains five (5) normal flags.

Scoring:

- Collecting a flag from your own side earns 10 points.
- Stealing a flag from your opponent's side earns 20 points.

Once a robot returns a flag to its checkpoint:

- A team member must manually remove the flag from the robot.
- The collected flag must be placed in the team's designated collection area.
- Only flags successfully removed and collected by hand will count towards the final score.

Main Flag:

Each team also has one Main Flag mounted on its flag post. If a team successfully captures the opponent's Main Flag and returns it to its own flag post, that team immediately wins the match regardless of the current score.

Bluetooth Round Rules:

- Only the magnet provided by the organizers may be used.
- No grabbing, clamping, or additional collection mechanisms are allowed.
- Robots may push each other; however, deliberate damage is strictly prohibited.
- Human contact with the robot is only permitted when removing a successfully collected flag at the checkpoint.

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Bluetooth Round Rules (Continued):

- Teams may not block the bridges intentionally for extended periods or obstruct the arena in an unsportsmanlike manner.
- Referees reserve the right to reset displaced arena elements if necessary.

Winning Conditions:

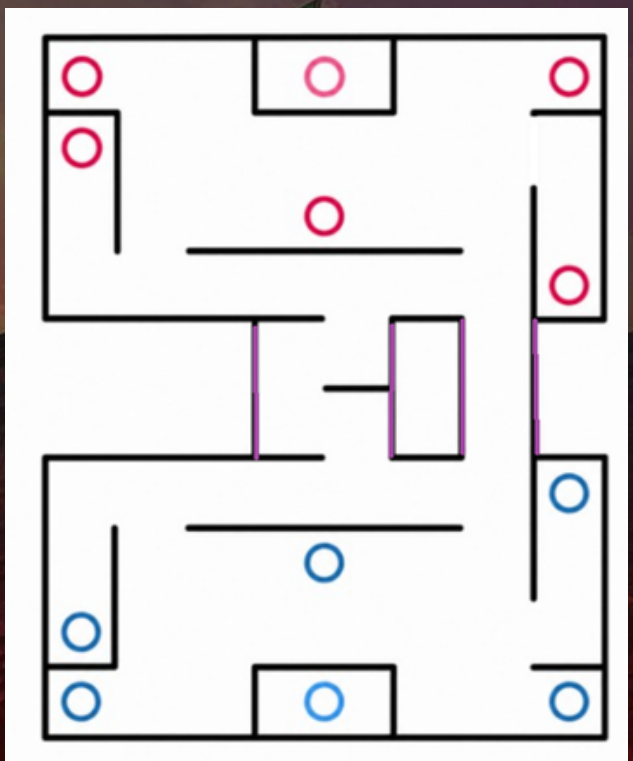
A team wins a match by:

- Successfully capturing the opponent's Main Flag

OR

- Having the highest score after the 5-minute match timer expires.

Track Features:



The Bridges are highlighted in Purple.
The Flags are highlighted in blue and red.

Scoring Summary:

Actions

- Collect flags from your own side
- Steal flags from opponent's side
- Capture the opponent's Main Flag

Points:

- 10 Points (OWN FLAG)
- 20 Points (OPPONENT FLAG)
- Instant Win (OPPONENT MAIN FLAG)

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Equipment to Bring:

Participants are advised to bring:

- Basic Toolkit (Paper cutter, Screwdrivers and Wirestripper)
- Laptop with Arduino IDE installed.
- USB Programming Cable
- Positive attitude and teamwork

Code of Conduct:

- All participants are expected to:
- Demonstrate good sportsmanship.
- Respect fellow competitors and event officials.
- Follow all safety instructions.
- Maintain a clean and organized workspace.
- Accept the judges' decisions respectfully.
- Basic tools may be provided to a team on judges' discretion.
- Any form of cheating, intentional interference, unsafe behavior, or misconduct may result in penalties or disqualification.

The Jury holds the right to make any changes to the robot kit, the track, the arena or the rules at any given time.